Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year 2020 – 2021
Semester Summer
Exam Date 8 / 9 / 2021

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELC 422) Digital Signal Processing — **Spec.:** Level 4 Electronic Eng.& Commu. Tech. Dep.
Examiners: Dr. Samir M Kamal & Dr. Sara M Hassan — **Time:** 2 hours

Number of Pages: One — Number of Questions: 4 — Attempt all Questions

Question 1 (15 points)

a- In A/D, in detail, explain the quantization process.
b- Find the relation between the frequency of the discrete time signal, the frequency of the continuous time signals and the sampling frequency.
c- For the signal $x(t) = 2\cos(100\pi t)$ to be sampled with sampling frequency F_S = 200Hz.
(i) Check the periodicity of x(t).
(ii) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.
(iii) Find the samples values of the first cycle in x(n) then plot them at the same figure with x(t).

Question 2 (15 points)

a- Define the following items: Correlation operation and Convolution operation.
b- If you apply the following input signal, x (n), x(n) = {3, 2, 0, 1} to a DTS represented by the following impulse response h(n), h(n) = {1, 2, 0, 1}, find the system output y(n).
c - Given two sets of samples, x_1 (n) = {3, 2, 0, 1} and x_2 (n) = {1, 2, 0, 1}:
(i) Find the cross- correlation between $x_1(n)$ and $x_2(n)$.
(ii) Plot the correlation function obtained in part (i) v.s the lag number j.
(iii) Find the autocorrelation function of the sequence $x_1(n)$.

Question 3 (15 points)

a- For LTIS define (using equations and/or figures) the following items: Causality and difference equation
b- Using the residue method, find the impulse response, h(n), of the system characterized by the following transfer function, $H(z)$.

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

c- For the transfer function given in part (b), H(z):
(i) Plot its pole – zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system, y (n) and
(v) Draw its realization diagram using the block diagram method.

Question 4 (15 points)

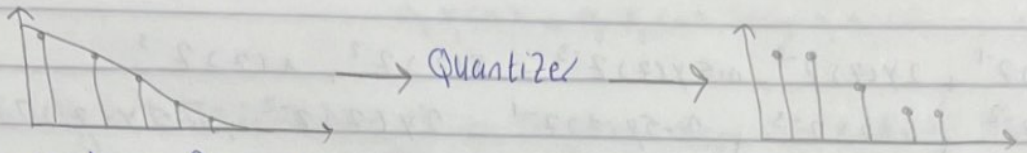
a- Compare between both filter types.
b- For the following sequence: x(n) = {1, 1, 0, 1}:
(i) Using the FFT algorithm find the frequency components of x(n) and check your results by calculating x(n) using the IDFT method.
(ii) Using your results, verify Parseval's theorem.
(iii) Plot the amplitude and the phase spectrums.

GOOD LUCK

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Q1:- a) - Converts continuous amplitude to discrete amplitude by representing each sample to the nearest level



Discrete time & Cont. Amp Discrete time & Amp.

Quantization error :- the difference between the unquantized sample & the quantized o/p $e_q(n) = x_a(n) - x(n)$ $e_q(n)$

b) $x(t) = A \sin(2\pi f t + \theta)$

$x(nts) = A \sin(2\pi f nts + \theta)$ $ts = 1/f_s$

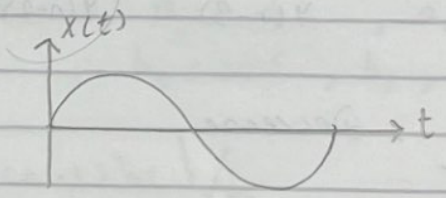
$x(nts) = A \sin\left(\frac{2\pi f}{f_s} n + \theta\right)$ $\frac{f}{f_s} \rightarrow F$

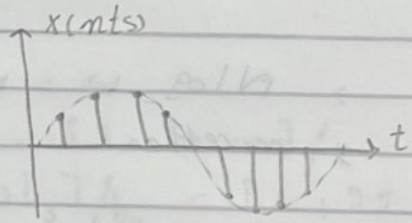
$= A \sin(2\pi F n + \theta)$

Digital freq

$F = \frac{f}{f_s}$ → Analog freq

f_s → sampling freq





A : Amplitude
f : freq = 1/tp (Hz)
tp : periodic time
θ : phase shift

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c) i) $x(t) = 2\cos(100\pi t)$, $f_s = 200\,Hz$

$x(t) = A\cos(2\pi f t)$, $2\pi f = 100\pi \rightarrow f = 50\,Hz$

$t_p = \frac{1}{50} = 20\,msec$

Replace each t by $t + t_p$

$x(t + t_p) = 2\cos(100\pi(t + t_p))$

$= 2\cos(100\pi t + 2\pi)$

$x(t + t_p) = 2\cos(100\pi t) = x(t)$

signal is periodic at $t_p = 20$ msec.

ii) * Replace each t by nts

$x(nts) = 2\cos(100\pi(nts))$ $\quad t_s = 1/f_s = 1/200 = 5\,msec$

$= 2\cos(100\pi \frac{n}{200})$

$x(nts) = 2\cos(\frac{\pi n}{2})$

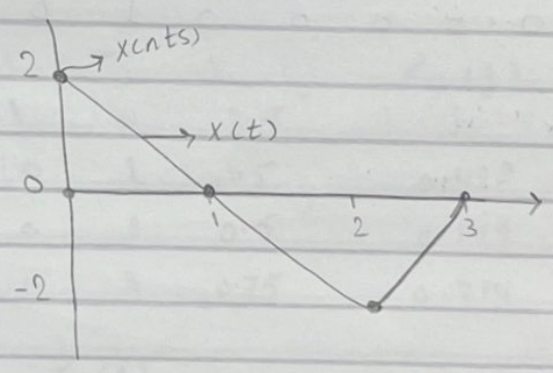
$F = \frac{f}{f_s} = \frac{50}{200} = \frac{1}{4} = \frac{K}{N_p}$, $N_p = 4$ samples

iii) $x(0) = 2$

$x(1) = 0$

$x(2) = -2$

$x(3) = 0$



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Q2 :-a) Correlation operation :- it measures the degree of similarity between 2 signals.

Convolution operation :- Describes how the i/p interact with the system to produce an o/p.

b) $x(n) = \{3, 2, 0, 1\} \rightarrow \{0,0,0,3,2,0,1\}$

$h(n) = \{1, 2, 0, 1\} \rightarrow \{1,0,2,1,0,0,0\}$

$N = 4+4-1 = 7$

								n	y(n)
x(n)	0	0	0	3	2	0	1		
	1	0	2	1	0	0	0	0	3
	0	1	0	2	1	0	0	1	8
x(n-m)	0	0	1	0	2	1	0	2	4
	0	0	0	1	0	2	1	3	4
	0	0	0	0	1	0	2	4	4
	0	0	0	0	0	1	0	5	0
	0	0	0	0	0	0	1	6	1

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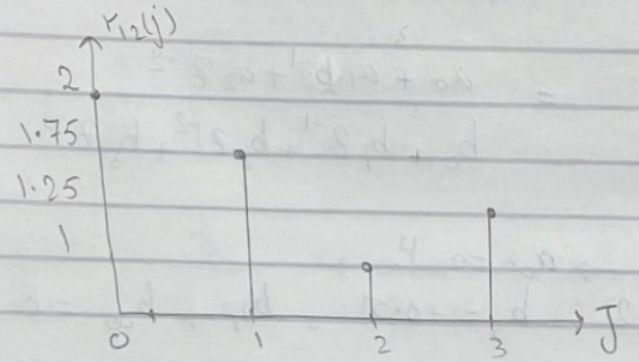
c) i) $x_1(n) = \{3, 2, 0, 1\}$, $x_2(n) = \{1, 2, 0, 1\}$

$N = 4 = N_{new}$ 'cyclic'

$x_1(n)$	3	2	0	1	j	$r_{12}(j)$	$\rho_{12}(j)$
	1	2	0	1	0	2	0.872
$x_2(n+j)$	2	0	1	1	1	1.75	0.763
	0	1	1	2	2	1	0.436
	1	1	2	0	3	1.25	0.545

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{4}\sqrt{14 * 6}} = \frac{r_{12}(j)}{2.291}$$

ii)



iii)

$x_1(n)$	3	2	0	1	j	$r_{11}(j)$	$\rho_{11}(j)$
	3	2	0	1	0	3.5	1
$n_1(n+j)$	2	0	1	0	1	1.5	0.428
	0	1	0	0	2	0.5	0.142
	1	0	0	0	3	0.75	0.214

$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{N}\sum_{0}^{N_{new}-1} x_1^2(n)} = \frac{r_{11}(j)}{\frac{1}{4} * 14} = \frac{r_{11}(j)}{3.5}$$

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Q3 :- a) Causality :- The i/p relationship for a LTIS is given by the convolution sum

$$Y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

Difference equation :- Represents the relation between the system response with an i/p in time domain.

b) By using power series :- $h(n) = [a_n - \sum_{i=1}^{n} h(n-i) b_i] / b_0$

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2} = \frac{z^3+z^2}{z^3-2.5z^2+2z-0.5} * \frac{z^{-3}}{z^{-3}}$$

$$H(z) = \frac{1+z^{-1}}{1-2.5z^{-1}+2z^{-2}-0.5z^{-3}} = \frac{a_0+a_1z^{-1}+a_2z^{-2}}{b_0+b_1z^{-1}+b_2z^{-2}+b_3z^{-3}}$$

$a_0 = 1$, $a_1 = 1$, $a_2 \to a_\infty = 0$

$b_0 = 1$, $b_1 = -2.5$, $b_2 = 2$, $b_3 = -0.5$, $b_4 \to b_\infty = 0$

$h(0) = a_0/b_0 = 1$

$h(1) = [a_1 - h(0)*b_1]/b_0 = 1 - (1*-2.5)/1 = 3.5$

$h(2) = [a_2 - h(1)b_1 - h(0)b_2]/b_0 = [0 - (3.5*-2.5) - (2)]/1$

$= 6.75$

$h(3) = [a_3 - h(2)b_1 - h(1)b_2 - h(0)b_3]/b_0$

$16.875 \quad -7 \quad -1$

$= 0 - (6.75*-2.5) - (3.5*2) - (1*-0.5)]/1$

$= 10.375$

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b) By using partial fraction

$$\frac{H(z)}{z} = \frac{z^2(z+1)}{z(z-0.5)(z-1)^2} = \frac{z(z+1)}{(z-0.5)(z-1)^2}$$

$$= \frac{A}{z-0.5} + \frac{c_2}{(z-1)^2} + \frac{c_1}{(z-1)}$$

$$A\Big|_{z=0.5} = \frac{z(z+1)}{(z-1)^2} = 3$$

$$c_2\Big|_{z=1} = \frac{z^2+z}{(z-0.5)} = \frac{z(z+1)}{(z-0.5)} = 4$$

$$c_1 = \frac{d}{dz}\left[\frac{(z-0.5)(2z+1)-(z^2+z)(1)}{(z-0.5)^2}\right]_{z=1} = -2$$

$$\frac{H(z)}{z} = \frac{3}{z-0.5} + \frac{4}{(z-1)^2} - \frac{2}{(z-1)}$$

$$H(z) = \frac{3z}{z-0.5} + \frac{4z}{(z-1)^2} - \frac{2z}{z-1}$$

$$h(n) = 3(0.5)^n + 4n - 2(1)^n$$

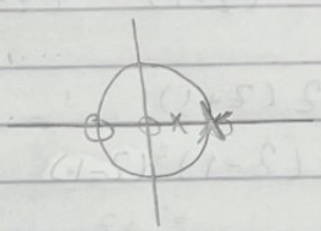
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o → Zeros
x → poles

c) i) poles :- $z = \{0.5, 1, 1\}$
Zeros : $z = \{0, 0, -1\}$

ii) Roc is the entire Z-plane except at $z = \{0.5, 1\}$

iii) system is marginally stable



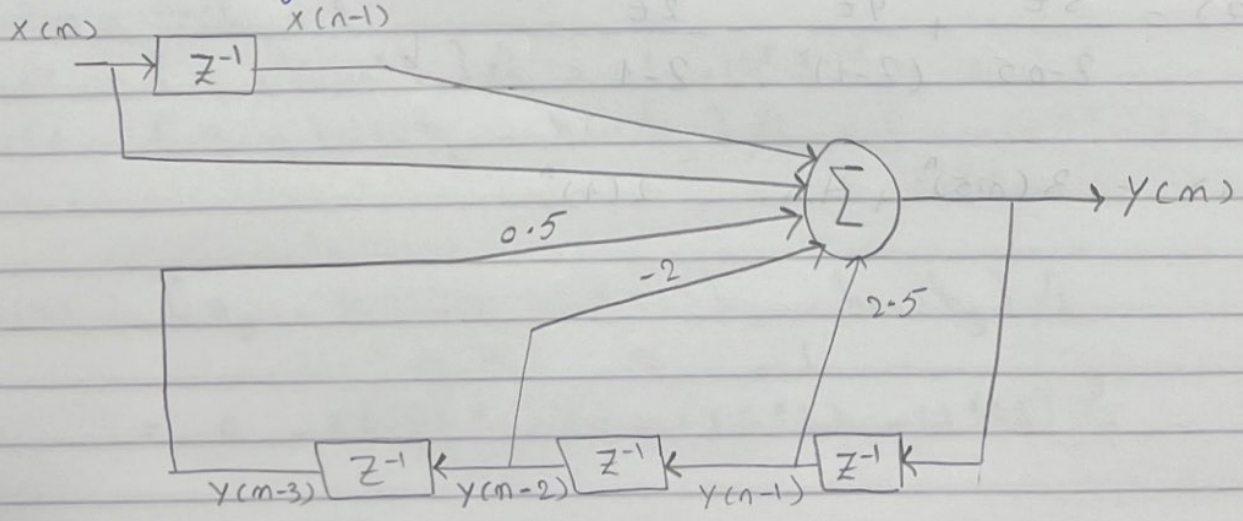
iv) $H(z) = \dfrac{1 + z^{-1}}{1 - 2.5z^{-1} + 2z^{-2} - 0.5z^{-3}} = \dfrac{Y(z)}{X(z)}$

$$Y(z) - 2.5Y(z)z^{-1} + 2Y(z)z^{-2} - 0.5Y(z)z^{-3} = X(z) + X(z)z^{-1}$$

$$Y(z) = X(z) + X(z)z^{-1} + 2.5Y(z)z^{-1} - 2Y(z)z^{-2} + 0.5Y(z)z^{-3}$$

$$y(n) = x(n) + x(n-1) + 2.5y(n-1) - 2y(n-2) + 0.5y(n-3)$$

v) Block Diagram



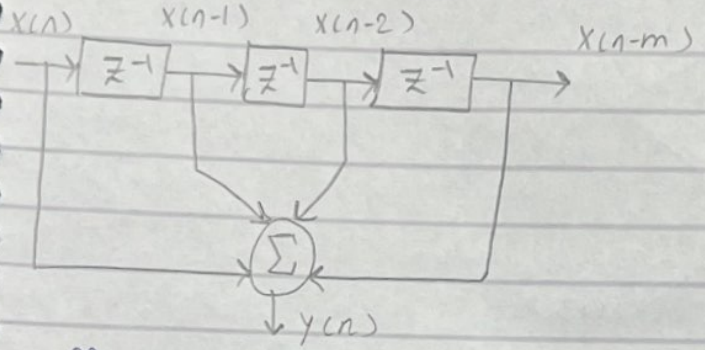
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a)

Q4 :- FIR

Finite impulse response

$$H(z) = \sum_{k=0}^{N-1} h(k)\, z^{-k}$$



Diff. eq :-

$$Y(n) = h(0)X(n) + h(1)X(n-1) + h(2)X(n-2) + h(m)X(n-m)$$

Low Delay & Accuracy

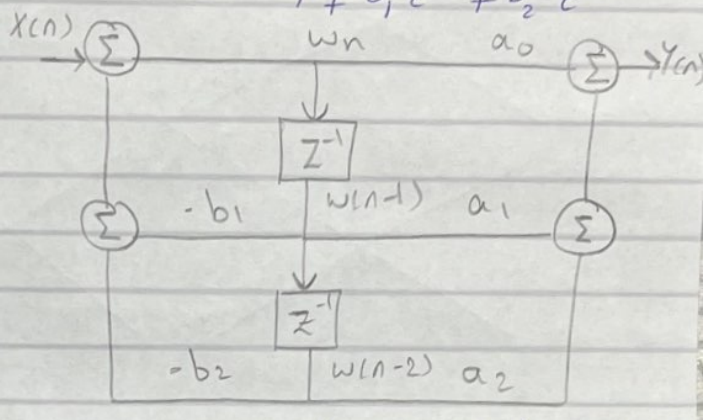
Alway stable 'No FB'

Realization

IIR

infinite impulse response

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$



$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$

$$y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$$

high Delay & Accuracy

may be stable may be not

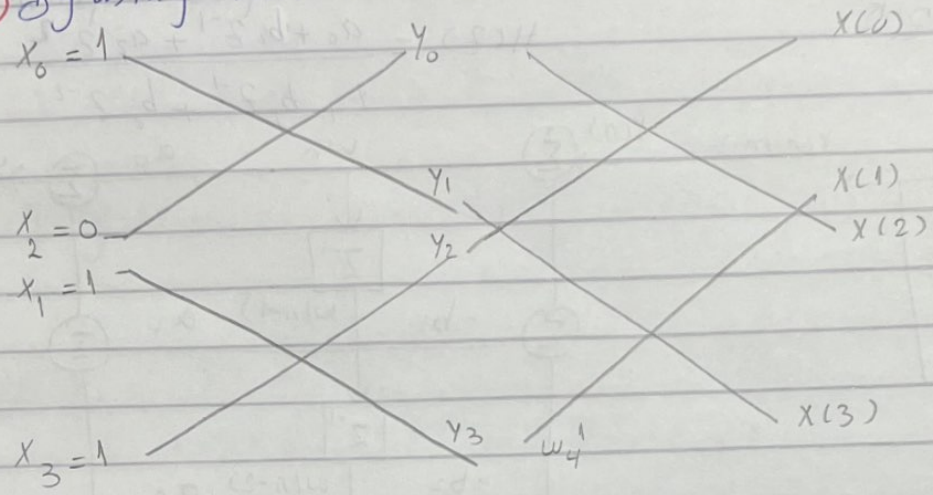
Realization

← Lattice →

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b) $x(n) = \{1, 1, 0, 1\}$

i) By using FFT :-



$w_4^1 = -j$

$y_0 = x_0 + x_2 = 1$
$y_1 = x_0 - x_2 = 1$
$y_2 = x_1 + x_3 = 2$
$y_3 = x_1 - x_3 = 0$

$X(0) = y_0 + y_2 = 3$
$X(1) = y_1 + (j) y_3 = 1 - 0 = 1$
$X(2) = y_0 - y_2 = -1$
$X(3) = y_1 - (j) y_3 = 1$

$X(K) = \{3, 1, -1, 1\}$

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By using IDFT $\quad x(k) = \{3, 1, -1, 1\}$

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} e^{j\frac{2\pi nk}{N}} = \frac{1}{4}\sum_{k=0}^{3} e^{j\frac{\pi nk}{2}}$$

$$x(n) = \frac{1}{4}\left[x(0) + x(1)\,e^{j\frac{\pi n}{2}} + x(2)\,e^{j\pi n} + x(3)\,e^{j\frac{\pi 3n}{2}}\right]$$

$$x(n) = \frac{1}{4}\left[3 + e^{j\frac{\pi n}{2}} - e^{j\pi n} + e^{j\frac{\pi 3n}{2}}\right]$$

$x(0) = 1$
$x(1) = 1$
$x(2) = 0$
$x(3) = 1$

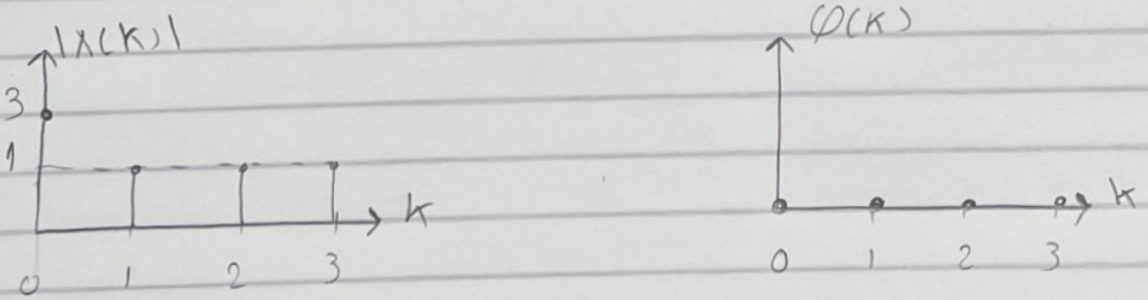
$x(n) = \{1, 1, 0, 1\}$

ii) Parseval's Theorem. $|X(k)| = \{3, 1, 1, 1\}$

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

$3 = 3 \rightarrow$ verified ✓

iii)



$\varphi(k) = \{0, 0, 0, 0\}$

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